APPENDIX I. PERIODIC TABLE OF THE ELEMENTS

1 H 1.008																	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.90	23 V 50.94	24 Cr 52.01	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.71	29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.91	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (97.91)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.4	47 Ag 107.87	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.90	54 Xe 131.30
55 Cs 132.91	56 Ba 137.34	57 - 71 *Ln	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.97	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.98	84 Po (208.98)	85 At (209.99)	86 Rn (222.02)
87 Fr (223.02)	88 Ra (226.03)	88 - 103 **An	104 Rf (267)	105 Db (268)	106 Sg (271)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113	114 Fl (289)	115	116 Lv (293)	117	118

*Lanthanides	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (144.91)	62 Sm 150.35	63 Eu 151.96	64 Gd 157.25	65 Tb 158.92	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**Actinides	89 Ac (227.03)	90 Th 232.04	91 Pa (231.04)	92 U 238.03	93 Np (237.05)	94 Pu (244.06)	95 Am (243.06)	96 Cm (247.07)	97 Bk (247.07)	98 Cf (251.08)	99 Es (252.08)	100 Fm (257.10)	101 Md (258.10)	102 No (259.10)	103 Lr (262)

APPENDIX II. QUANTITIES AND UNITS

Quantity	Symbol	Unit	Symbol	Dimensions (within brackets), derived units etc.
Length	l	meter	m	1 fermi = 10^{-15} m = 1 fm; 1 $\mu = 10^{-6}$ m; 1 Å = 10^{-10} m
Mass	m	kilogram	kg	Basic SI unit; 1 ton (t) = 10^3 kg; 1 kg = 10^3 g (gram)
Time	t	second	s	" ; 1 (ephem.) year (y or a) = 365.24 days (d) =
Electric current	I	ampere	A	" ; = 8765.8 hours (h) = 3.1557×10^7 s
Thermodynamic temperature	T	kelvin	K	" ; $t_C = t_K - 273.15$, t_C = temp. in degree Celsius, °C
Amount of substance	n, ν	mole	mol	" ; molarity (mol/l) = $m_a M^{-1} V^{-1}$, m_a = mass of pure
Luminous intensity	I^*	candela	cd	" ; substance;
Atomic (molecular) weight	M	atomic mass unit	u	$M = m_i \times 10^3 \times N_A$, m_i mass of atom (molecule); 1 mole = M g
Volume	V	cubic meter	m^3	1 $m^3 = 10^3$ liter (l, dm^3, L), 1 l = 10^3 milliliters (ml), 1 ml = 1 cm^3
Density (mass per unit volume)	ρ		kg m^{-3}	1 g $cm^{-3} = 10^3$ kg m^{-3}
Pressure	P	pascal	Pa	[Pa = Nm^{-2} = kg s^{-2} m^{-1}]; 1 atm = 1.013×10^5 Pa; 1 bar = 10^5 Pa
Energy	E	joule	J	[J = Ws = N m = kg m^2 s^{-2}] 1 torr = 133.3 Pa
Frequency	f, ν	hertz	Hz	[Hz = s^{-1}]
Force	F	newton	N	[N = kg m s^{-2}]; 1 dyne = 10^{-5} N; 1 kp = 9.8067 N
Angle, flat		radian	rad	1 rad = 57.30°; full circle 360° = 2π rad
" , space		steradian	sr	full space angle = 4π sr
Power	P	watt	W	[W = J s^{-1} = N m s^{-1} = kg m^2 s^{-3}]
Angular frequency	ω		rad s^{-1}	
Electric potential (voltage)	U	volt	V	[V = W A^{-1} = kg m^2 s^{-3} A^{-1}]
resistance	R	ohm	Ω	[Ω = V A^{-1} = W A^{-2} = kg m^2 s^{-3} A^{-2}]
charge	q	coulomb	C	[C = A s]; 1 C = 0.1 **c** statcoulomb (esu)
capacitance	C	farad	F	[F = C V^{-1} = s A^2 W^{-1} = s^4 A^2 kg^{-1} m^{-2}]
Magnetic inductance	L^*	henry	H	[H = Wb A^{-1} = V s A^{-1} = kg m^2 s^{-2} A^{-2}]
induction (flux density)	B	tesla	T	[T = Wb m^{-2} = V s m^{-2} = kg s^{-2} A^{-1}]; 1 gauss = 10^{-4} T
flux	$\phi_B{}^*$	weber	Wb	[Wb = V s = W s A^{-1} = kg m^2 s^{-2} A^{-1}]
Radioactivity	A	becquerel	Bq	[Bq = (radioactive events) s^{-1}]; 1 curie (Ci) = 3.7×10^{10} Bq
Radiation exposure		kerma		[C kg^{-1} = A s kg^{-1}]
Radiation dose, absorbed	D	gray	Gy	[Gy = J kg^{-1} = m^2 s^{-2}]; 1 Gy = 100 rad **
Radiation dose, equivalent	H	sievert	Sv	[Sv = J kg^{-1} = m^2 s^{-2}]; 1 Sv = 100 rem **

* Not used in this text. ** Old radiation dose unit (see §7.3).

APPENDIX III. FUNDAMENTAL CONSTANTS

Quantity	Symbol	Value	SI-unit	Auxiliary value
Speed of light in vacuum	$\mathbf{c}$	299 792 458	m/s	Basic SI unit (exact)
Elementary charge	$\mathbf{e}$	$1.602\,177\,3\times10^{-19}$	C	
Planck constant	$\mathbf{h}$	$6.626\,076\times10^{-34}$	J s	$= 4.135\,669\times10^{-15}$ eV s; $\hbar = \mathbf{h}/2\pi = 1.054\,572\,7\times10^{-34}$ J s
Avogadro constant	$\mathbf{N_A}$	$6.022\,137\times10^{23}$	mol^{-1}	
Atomic mass unit	u	$1.660\,540\times10^{-27}$	kg	= 931.494 3 MeV; mass of ^{12}C = 12 u
Electron rest mass	$\mathbf{m_e}$	$9.109\,390\times10^{-31}$	kg	$\mathbf{M_e} = \mathbf{N_A}\times\mathbf{m_e} = 5.485\,799\,0\times10^{-4}$ u = 0.510 999 1 MeV
Proton rest mass	$\mathbf{m_p}$	$1.672\,623\times10^{-27}$	kg	$\mathbf{M_p} = \mathbf{N_A}\times\mathbf{m_p} = 1.007\,276\,47$ u = 938.272 3 MeV
Neutron rest mass	$\mathbf{m_n}$	$1.674\,929\times10^{-27}$	kg	$\mathbf{M_n} = \mathbf{N_A}\times\mathbf{m_n} = 1.008\,664\,90$ u = 939.565 6 MeV
Faraday constant	$\mathbf{F}$	96 485.31	$C\,mol^{-1}$	$= \mathbf{N_A}\times\mathbf{e}$
Rydberg constant	$\mathbf{R_\infty}$	10 973 731.53	m^{-1}	$\mathbf{R_\infty}\times\mathbf{h}\times\mathbf{c}$ = 13.605 698 eV
Inverse fine structure constant	$\boldsymbol{\alpha}^{-1}$	137.035 990		$= \mu^0\times\mathbf{c}\times\mathbf{e}^2/2\times\mathbf{h}$; μ^0 (permeability of vacuum) $=4\pi\times10^{-7}$ H m^{-1}
Bohr radius	$\mathbf{a_0}$	$0.529\,177\,25\times10^{-10}$	m	$= \boldsymbol{\alpha}/4\pi\,\mathbf{R_\infty}$
Electron magnetic moment	μ_e	$-9.284\,770\times10^{-24}$	$J\,T^{-1}$	
Proton magnetic moment	μ_p	$1.410\,607\,6\times10^{-26}$	$J\,T^{-1}$	
Neutron magnetic moment	μ_n	$-0.966\,237\,0\times10^{-26}$	$J\,T^{-1}$	
Bohr magneton	μ_B	$9.274\,015\times10^{-24}$	$J\,T^{-1}$	$= \mathbf{e}\times\hbar/2\times\mathbf{m_e}$; $(1\,J\,T^{-1} = 10^3\,erg\,gauss^{-1})$
Nuclear magneton	μ_N	$5.050\,787\times10^{-27}$	$J\,T^{-1}$	$= \mathbf{e}\times\hbar/2\times\mathbf{m_p}$
Molar gas constant	$\mathbf{R}$	8.314 51	$J\,mol^{-1}\,K^{-1}$	$= 0.082\,06$ l atm $mol^{-1}\,K^{-1}$
Molar volume of ideal gas at STP	$\mathbf{V_0}$	$22.414\,1\times10^{-3}$	$m^3\,mol^{-1}$	$= \mathbf{R}\,T_0/p_0$; $T_0 = 273.15$ K, $p_0 = 1$ atm = 101 325 Pa
Boltzmann constant	$\mathbf{k}$	$1.380\,66\times10^{-23}$	$J\,K^{-1}$	$= \mathbf{R}/\mathbf{N_A} = 8.617\,39\times10^{-5}$ eV K^{-1}; $1/\mathbf{k} = 11\,604.4$ K eV^{-1}
Gravitational constant	$\mathbf{G}$	$6.672\,0\times10^{-11}$	$N\,m^2\,kg^{-2}$	
Acceleration of gravity at sea level	$\mathbf{g}$	9.806 65	$m\,s^{-2}$	(at 45° latitude)

APPENDIX IV. ENERGY CONVERSION FACTORS (CODATA 1986)

Joule (J)	Kilowatthour (kWh)	Atomic mass unit (u)	Kilocalorie (kcal)	Electronvolt (eV)
1	$2.777\,78\times10^{-7}$	$6.700\,53\times10^{9}$	$2.388\,46\times10^{-4}$	$6.241\,46\times10^{18}$
$3.600\,00\times10^{6}$	1	$2.412\,19\times10^{16}$	$8.598\,46\times10^{2}$	$2.246\,93\times10^{25}$
$1.492\,419\times10^{-10}$	$4.145\,61\times10^{-17}$	1	$3.564\,58\times10^{-14}$	$9.314\,943\times10^{8}$
$4.186\,80\times10^{3}$	$1.163\,00\times10^{-3}$	$2.805\,38\times10^{13}$	1	$2.613\,20\times10^{22}$
$1.602\,177\times10^{-19}$	$4.450\,53\times10^{-26}$	$1.073\,544\times10^{-9}$	$3.826\,77\times10^{-23}$	1

1 eV = $1.602\,177\times10^{-19}$ J; 1 eV $atom^{-1}$ = 23.045 0 kcal mol^{-1} = 96.485 kJ mol^{-1}
1 Q = 10^{18} Btu; 1 Btu (British thermal unit) = 1.055 06 kJ; 1 hp (horse power) = 0.746 kW
1 toe (ton oil equivalent) = 10 Gcal = 11.63 MWh = 41.87 GJ
1 ton hard coal (tce) = 0.65 toe = 27.2 GJ; 1000 m^3 natural gas = 0.80 toe
1 g ^{235}U fissioned at 200 MeV/fission = 82.11 GJ = 0.95 MWd (heat)
1 erg (dyne cm) = 1.000×10^{-7} J
Energy-wavelength product ($\Delta E\,\lambda$) = 12 398.5 eV Å

Prefixes for powers of ten			Some numerical values		Some English measures	
E	exa	10^{18}	e	2.718 28	1 inch	0.025 4 m
P	peta	10^{15}	log e	0.434 29	1 (statue) mile	1 609.34 m
T	tera	10^{12}	ln 2	0.693 15	1 (int.) nautical mile	1 852 m
G	giga	10^{9}	ln 10	2.302 59	1 (US liq.) gallon	0.003785 m^3
M	mega	10^{6}	ln 2/ln 10	0.301 03	1 barrel	0.159 0 m^3
k	kilo	10^{3}	π	3.141 59	1 cubic foot	0.028 32 m^3
h	hecto	10^{2}	ln a = ln 10 × log a		1 pound (mass)	0.4536 kg
d	deci	10^{-1}				
c	centi	10^{-2}				
m	milli	10^{-3}				
μ	micro	10^{-6}				
n	nano	10^{-9}				
p	pico	10^{-12}				
f	femto	10^{-15}				
a	atto	10^{-18}				